

CONSTRUCTING DEGREE 4 VERTEX RIGID-FOLDABLE THICK ORIGAMI BY VOLUME TRIMMING

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1. INTRODUCTION

Origami often relies on the assumption of using ideal zero-thickness paper or film. This allows for folding various crease patterns like flat-foldable degree 4 vertices. However, there are some challenges when using rigid material to fold, such as metal, wood, and plastic. Unlike ideal zero-thickness paper, rigid panels have physical thickness. The panels will collide into each other along the crease line and not fold properly due to the adding thickness. During folding, especially towards smaller dihedral angles, the material on either side of a crease interferes the folding motion even potentially damages the material. This physical restriction means that the simple angle relationships (like Kawasaki's theorem for flat-foldability) cannot be applied directly to the thick folding. The material thickness fundamentally alters the geometry at the hinges. Therefore, techniques such as volume trimming are essential to create the necessary space for panels to fold without collision.

Many thick material folding techniques exist. In this paper, I will focus on volume trimming technique to overcome the challenge of thick origami. Volume trimming, first proposed by Tachi, is a method for constructing hinges by removing material near the fold lines. While simple to fabricate, the original method had limitations, particularly delamination, where stiff layers detach from the folding layer due to external forces. To address delamination in the folding, Jason Ku developed **axially varying volume trimming** to improve the strength of the hinge.

I will demonstrate volume trimming, the improvement by axial variation, and its application to construct degree 4 vertices as shown in the Figure 1 using laser-cut wood and Acrylic.

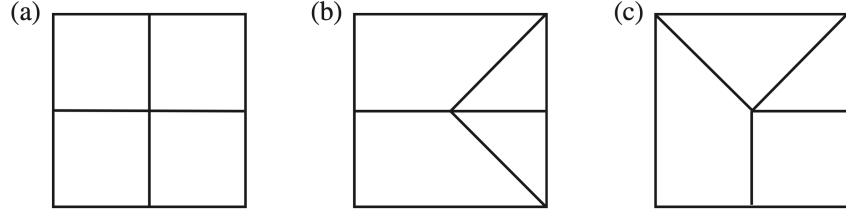


FIGURE 1. Three different degree-4 flat vertex folds, by Thomas Hull [4]

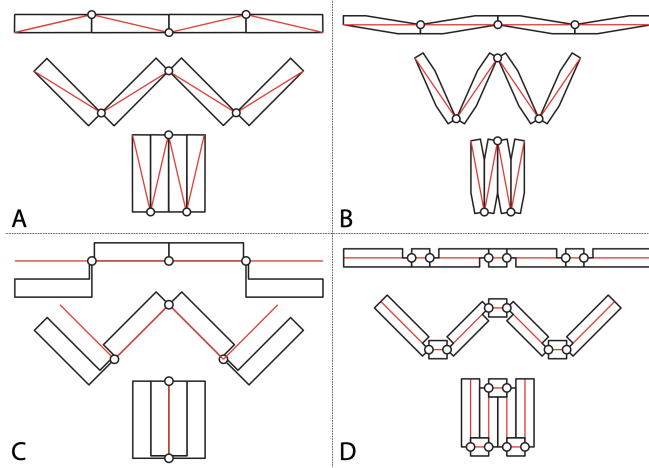


FIGURE 2. Existing thick folding techniques (A) hinge shift, (B) volume trimming, (C) offset panel, and (D) offset crease, by Jason Ku. [1]

2. EXISTING THICK FOLDING TECHNIQUES

2.1. **A: Hinge Shift.** This method shifts hinges out of the plane to adapt the material thickness. Figure 2

2.2. **B: Volume Trimming.**

Definition 2.1. Volume trimming is a technique to allow rigid folding of thick materials by removing interfering volumes of material near the hinge lines. (Figure 5)

2.3. **C: Offset panel.** This method keeps the hinge in the folding plane but moves the panel away from the folding plane. Figure 2

2.4. **D: Offset Crease.** This method splits each crease into two; allows full motion but adds degrees of freedom. Figure 2

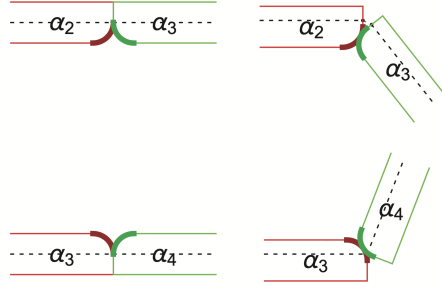


FIGURE 3. (E) Rolling Contact, by Robert J. Lang [3]

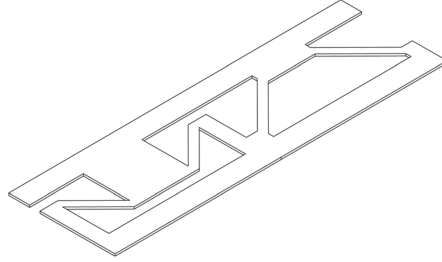


FIGURE 4. (F) Strained Joint, by Isaac L. Delimont [2]

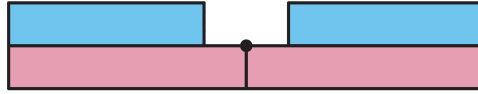


FIGURE 5. Diagram of simple volume trimming, J. S. Ku [1]

2.5. E: Rolling Contacts. Hinges are constructed using elements that roll against each other, similar to a Jacob's Ladder toy. This method allows the hinge axis to shift based on the shape of the rolling surfaces. Figure 3

2.6. F: Strained Joint. Strained Join creates hinges in a single, continuous sheet of thick material by removing some material along the fold line. Figure 4

3. CHALLENGES IN FOLDING THICK MATERIALS AND ORIGINAL VOLUME TRIMMING

Folding thick material introduces challenges compared to thin paper. When thick panel folds, their inner corners collide unless material is

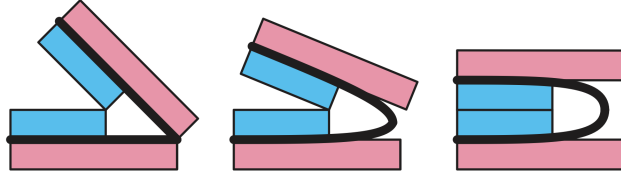


FIGURE 6. Delamination when the folding layer is not contained(sandwiched) between thick panels, by J. S. Ku [1]

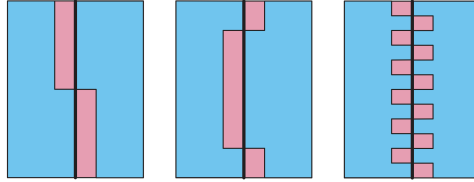


FIGURE 7. Top view of piecewise constant axial variation in volume trimming for one, two, and many alterations, by J. S. Ku [1]

removed or the hinge is shifted. The hinge shifted (Figure 2 A) works fine with 1-D pattern but is difficult for 2-D.

Tachi's original volume trimming method also has a limitation called delamination as shown in Figure 6. The reason is that this uniform removal along the hinge line leaves the folding layer unsupported. In other words, it is not contained.

Definition 3.1. A folding layer is said to be **contained** when it sandwiched fully by both sides.

4. AXIALLY VARYING VOLUME TRIMMING

To address this issue, Jason Ku introduced axially varying volume trimming with teeth pattern that adds alteration along the hinge axis. The main idea is to ensure the flexible folding layer remains contained by the rigid thick material and keeping panel close to the fold line. The sufficient retaining material on both sides of the fold line helps to maintain the strength of the hinge and panel stiffness.

4.1. Small Fold Angles ($0^\circ \leq \gamma \leq 90^\circ$). For small fold angles, containment can be achieved by removing material of only one of the inner sides of the hinge. To ensure containment on both sides of the hinge, the trimmed side's removal is alternated along the hinge line. This

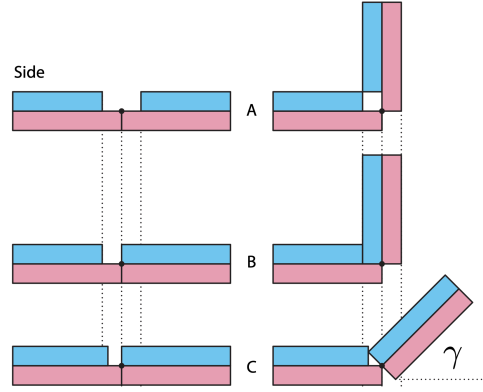


FIGURE 8. Small Fold Angles ($0^\circ \leq \gamma \leq 90^\circ$), by J. S. Ku [1].

constructs interlocking “teeth” pattern as shown in Figure 7. A single alternation could still have delamination so two or more alternations are more ideal for stability purpose.

To determine the trimming distance for small angle, consider the cross-section shown in Figure 8. Suppose the panel thickness be t and fold angle be γ , we need to trimming the material at the distance of d from the fold line. Ku [1] claims that trimming distance $d = t \cot \gamma$. The side opposite the angle γ is the thickness t of the folding panel, and the side adjacent to γ is the horizontal distance d we need to trim. $\tan(\gamma) = \frac{\text{opposite}}{\text{adjacent}} = \frac{t}{d}$. So, we get $d = t / \tan(\gamma)$. We know that $\cot(\gamma) = 1 / \tan(\gamma)$. Therefore, the distance is $d = t \cot(\gamma)$

4.2. Large Fold Angles ($90^\circ < \gamma \leq 180^\circ$). For larger angles, simple trimming improvements don’t help to contain the folding layer. Trimming must be done on both inner sides. The corresponding trimming distances are shown in Figure 9. Compared with simple small angle fold, the distance formula for horizontal pink panel becomes $d = t \cot(\pi - \gamma)$.

5. APPLICATION IN DEGREE 4 VERTICES

Definition 5.1. A **degree 4 vertex** in an origami crease pattern is a point where exactly four crease lines meet.

Degree 4 vertices are fundamental components in origami. Applying axially varying volume trimming allows to use thick rigid material to fold these vertices.

This project focuses on three specific flat-foldable vertices:

- Bird-foot Vertex ($\alpha = 45^\circ, \beta = 135^\circ$)

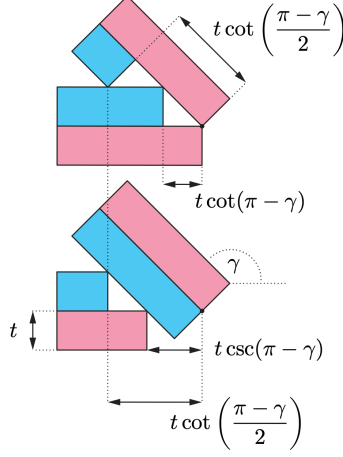


FIGURE 9. Distances for various trimming when applying large angle fold ($90^\circ < \gamma \leq 180^\circ$), by J. S. Ku [1].

- Vertex of ($\alpha = 45^\circ, \beta = 90^\circ$)
- Four Quarters Vertex ($\alpha = 90^\circ, \beta = 90^\circ$)

, where α is the smallest angle and its adjacent angle is β .

The cutting patterns below were generated using the “Four Crease Hinge Layout” [5] web tool developed by Ku. These patterns incorporate the axially varying volume trimming with teeth pattern that allows fold angle up to 90° . The generated patterns include the necessary space near the central vertex to prevent collisions during folding. These features are designed for 2-D fabrication like laser cutter on wood or Acrylic sheets, with a flexible material as folding layer.

6. CONCLUSION

Transitioning origami from thin paper to thick material raises challenges of material collision and hinge delamination. Volume trimming provides a feasible solution by removing interfering material. The improvement of axially varying volume trimming using teeth alternation significantly prevents delamination by retaining enough material along the fold axis to contain the middle flexible folding layer.

By employing axially varying volume trimming, origami pattern under zero-thickness assumption can be constructed from rigid thick material, providing more possible solutions in engineering. The fabrication example of these vertices using wood and Acrylic validates the effectiveness of the axially varying trimming method.

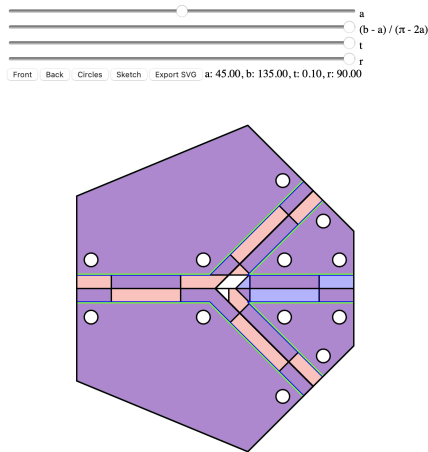


FIGURE 10. Bird-foot Vertex ($\alpha = 45^\circ$, $\beta = 135^\circ$) cutting pattern using axially varying volume trimming, generated by the web application [5].

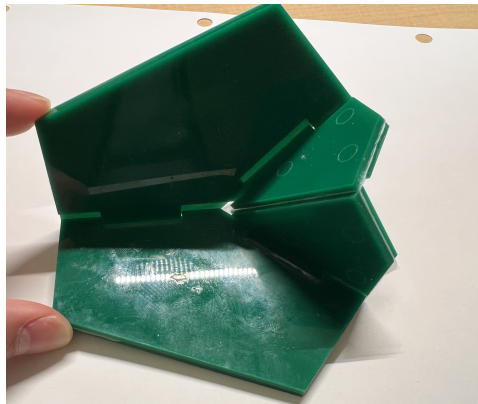


FIGURE 11. Fabrication example of Bird-foot pattern, corresponding to Figure 10 cutting pattern

7. ACKNOWLEDGMENTS

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REFERENCES

- [1] J. S. Ku, Folding thick materials using axially varying volume trimming, in *Proceedings of the ASME 2017 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE 2017)*, DETC2017-67577, Cleveland, USA (2017).

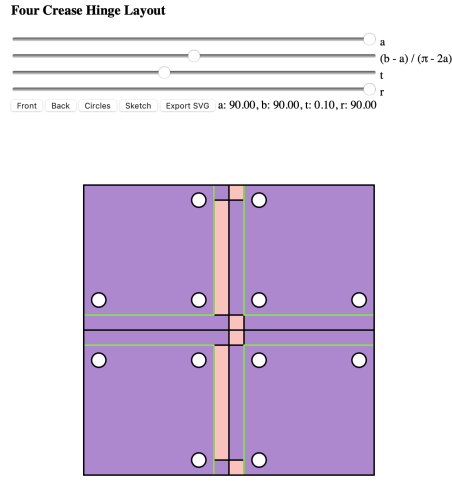


FIGURE 12. Four Quarters Vertex ($\alpha = 90^\circ, \beta = 90^\circ$) cutting pattern using axially varying volume trimming, generated by the web application [5].

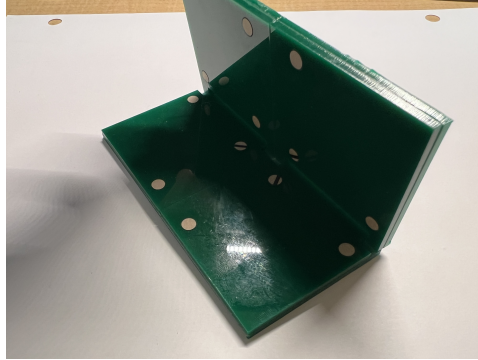


FIGURE 13. Fabrication example of Four Quarters Vertex pattern, corresponding to Figure 12 cutting pattern

- [2] I. L. Delimont, S. P. Magleby, and L. L. Howell, *A Family of Dual-Segment Compliant Joints Suitable for Use as Surrogate Folds*, Journal of Mechanical Design, Vol. 137, No. 9 (2015), pp. 092302-1–092302-10, ASME. DOI: 10.1115/1.4030875.
- [3] R. J. Lang, T. Nelson, S. Magleby, and L. Howell, *Thick Rigidly Foldable Origami Mechanisms Based on Synchronized Offset Rolling Contact Elements*, Proceedings of the ASME 2016 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE 2016), DETC2016-59747, Charlotte, North Carolina, USA, August 21–24, 2016. ASME. URL: <http://proceedings.asmedigitalcollection.asme.org/pdfaccess.ashx?url=/data/conferences/asmep/90696/>

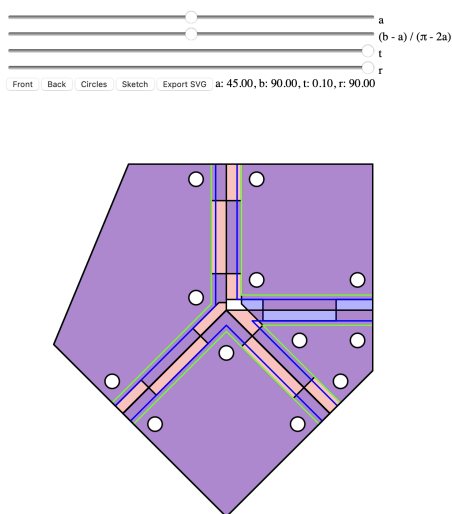


FIGURE 14. Vertex of $(\alpha = 45^\circ, \beta = 90^\circ)$ cutting pattern using axially varying volume trimming, generated by the web application [5].



FIGURE 15. Fabrication example of $(\alpha = 45^\circ, \beta = 90^\circ)$ Vertex pattern, corresponding to Figure 12 cutting pattern.

- [4] T. C. Hull, *Origametry: Mathematical Methods in Paper Folding*, Cambridge University Press, Cambridge, UK (2020), ISBN: 9781108478724. DOI: 10.1017/9781108778633.
- [5] J. S. Ku, *Four Crease Hinge Layout*, web tool, accessed April 23, 2025. URL: <http://jasonku.mit.edu/four-crease-hinge-layout/>.

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